

CUSTOM AI TOOL CONCEPT

The most transformative professionals in any field are not just users of tools — they are toolmakers. This optional assignment challenges you to step into that role by identifying a real bottleneck or pain point in your architectural workflow and conceptualizing an AI-driven solution to address it. The best proposals will be grounded in firsthand frustration: a repetitive task you've faced, an information gap that slows decisions, or a communication challenge between disciplines that AI could help bridge.

Your proposal should be specific, practical, and architecturally grounded. Generic proposals like "an AI that makes better renderings" will not score well. Strong proposals define a precise problem, articulate the data inputs the tool would require, describe the expected output in concrete terms, and explain clearly how the tool would integrate into and enhance the architectural design process. Think of this as the conceptual seed of a startup pitch — it should be convincing enough that a developer could begin building from your description alone.



PROBLEM DEFINITION

Clearly identify the specific bottleneck or challenge in the architectural workflow that your tool is designed to solve.



DATA INPUTS

Describe precisely what data or information the tool requires to function — drawings, schedules, specifications, sensor data, etc.



EXPECTED OUTPUT

Define what the tool generates — reports, visualizations, schedules, design options — and how it would be used by the architect.



PROCESS INTEGRATION

Explain how this tool specifically supports and enhances the architectural design process at a critical stage of the project lifecycle.

Criteria	Points
Clear identification of the problem and how the tool supports the process	4 pts
Logical mapping of required inputs to expected outputs	6 pts
Total	10 pts



This is an optional assignment worth up to 10 bonus points. Even if custom tool AI mapping is new to you, attempting this assignment is a valuable learning experience — and partial credit is awarded for honest effort and documented exploration.

PARAMETRIC + AI DESIGN STRATEGY

This optional assignment invites you to explore one of the most intellectually rich frontiers in contemporary architectural practice: the intersection of rule-based parametric design and artificial intelligence. Parametric scripting allows designers to encode logic directly into geometry, creating systems that respond dynamically to changing inputs — whether those inputs are environmental data, client requirements, or structural constraints. When combined with AI, these systems can become adaptive, generative, and deeply responsive to real-world complexity.

Your task is to develop a parametric script that is directly responsive to your client's needs and remains narratively aligned with your pavilion project. The script should not feel like a separate technical exercise — it should feel like a natural extension of your design thinking, encoding the same spatial and material logic that drives your hand drawings and AI-generated images. Students who complete this assignment will demonstrate a significant advancement in their technical toolkit and their ability to think systematically about design.



FUNCTIONAL LOGIC

Your script must demonstrate clear, working parametric logic — not just a visual result, but an underlying rule system that drives the geometry.



CLIENT RESPONSIVENESS

The parametric output must accurately respond to defined client requirements and align with the overall project narrative.

Criteria	Points
Script demonstrates functional parametric logic	5 pts
Output responds accurately to client requirements and narrative alignment	5 pts
Total	10 pts



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